Particle Interactions and Conservation Laws

For each of the following interactions:

- State whether the interaction is **valid** or **invalid**.
- For **exotic particles**, the **quark composition is provided** so you can determine whether the particle is a **meson or baryon** (and hence its **baryon number**) and work out its **charge**.
- If invalid, identify which conservation law is violated:
  - Charge
  - Lepton number
  - Baryon number

Only these three conservation laws should be considered.

1. $$n \rightarrow p + e^- + \bar{\nu}_e$$

2. $$p + e^- \rightarrow n$$

3. $$\pi^+ \rightarrow \mu^+ + \nu_\mu$$

4. $$\mu^- \rightarrow e^- + \nu_e$$

5.

$$p \to e^+ + \gamma$$

6. K^+ is a particle with quark content $u\bar{s}$.

$$K^+ \to \pi^+ + \pi^0$$

7. Δ^{++} is a particle with quark content uuu.

$$\Delta^{++} \to p + \pi^+$$

8. Λ^0 is a particle with quark content uds.

$$\Lambda^0 \to p + \pi^-$$

9. K^0 is a particle with quark content $d\bar{s}$.

$$K^0 \to e^- + \pi^+$$

10.

$$p + \bar{p} \to \pi^+ + \pi^- + \pi^0$$

11.

$$n \to p + e^-$$

12.

$$\mu^+ \rightarrow e^+ + \bar{\nu}_\mu$$

13.

$$e^- + e^+ \rightarrow \gamma$$

14.

$$\pi^- \rightarrow \mu^- + \bar{\nu}_\mu$$

15.

$$\pi^0 \rightarrow e^- + e^+$$

16. K^- is a particle with quark content $s\bar{u}$.

$$K^- \rightarrow \mu^- + \bar{\nu}_\mu$$

17.

$$p + n \rightarrow p + p + e^- + \bar{\nu}_e$$

18.

$$n + \bar{\nu}_e \rightarrow p + e^-$$

19.

$$\mu^- \rightarrow e^- + \gamma$$

20.

$$e^- \rightarrow e^- + \gamma$$

21. Σ^+ is a particle with quark content uus.

$$\Sigma^+ \rightarrow p + \pi^0$$

22. Σ^- is a particle with quark content dds.

$$\Sigma^- \rightarrow n + \pi^-$$

23. Ξ^0 is a particle with quark content uss.

$$\Xi^0 \rightarrow \Lambda^0 + \pi^0$$

24. Λ^0 is a particle with quark content uds.

$$\Lambda^0 \rightarrow p + e^- + \bar{\nu}_e$$

25.

$$p + p \rightarrow p + p + \pi^0$$

26.

$$n + n \rightarrow p + p + e^- + e^-$$

27.

$$\mu^+ + \mu^- \rightarrow e^+ + e^-$$

28.

$$e^- + p \rightarrow e^- + p + \pi^0$$

29. D^+ is a particle with quark content $c\bar{d}$.

$$D^+ \rightarrow K^0 + \pi^+$$

30. D^0 is a particle with quark content $c\bar{u}$.

$$D^0 \rightarrow K^- + \pi^+$$

31.

$$p \rightarrow p + \pi^0$$

32.

$$n \rightarrow n + \gamma$$

33.

$$\pi^+ \rightarrow e^+ + \nu_e$$

34.

$$\mu^- \rightarrow e^- + \bar{\nu}_e$$

35.

$$e^- + \bar{\nu}_e \rightarrow \mu^-$$

36.

$$p + \bar{p} \rightarrow \mu^+ + \mu^-$$

37. Ω^- is a particle with quark content sss.

$$\Omega^- \rightarrow \Lambda^0 + K^-$$

38. Ω^- is a particle with quark content sss.

$$\Omega^- \rightarrow p + \pi^-$$

39.

$$n + p \rightarrow p + p$$

40.

$$n + \pi^+ \rightarrow p$$

41.

$$e^- + \gamma \to e^-$$

42.

$$\pi^0 \to \gamma + \gamma$$

43. B^+ is a particle with quark content $u\bar{b}$.

$$B^+ \to \pi^+ + \pi^0$$

44. B^0 is a particle with quark content $d\bar{b}$.

$$B^0 \to p + \bar{p}$$

45.

$$p \to n + e^+ + \nu_e$$

46.

$$n \to p + e^-$$

47.

$$\mu^- + p \to n + \nu_\mu$$

48.

$$e^- + e^+ \rightarrow \mu^- + \mu^+$$

49. Σ^0 is a particle with quark content uds.

$$\Sigma^0 \rightarrow \Lambda^0 + \gamma$$

50.

$$p + \pi^- \rightarrow n$$

Answers

1. Valid
2. Invalid — Lepton number
3. Valid
4. Invalid — Lepton number
5. Invalid — Baryon number
6. Valid
7. Valid
8. Valid
9. Invalid — Lepton number
10. Valid
11. Invalid — Lepton number
12. Invalid — Lepton number
13. Valid
14. Valid
15. Valid
16. Valid
17. Valid
18. Invalid — Lepton number
19. Invalid — Lepton number
20. Valid
21. Valid
22. Valid
23. Valid
24. Valid
25. Valid
26. Invalid — Lepton number
27. Valid
28. Valid
29. Valid

30. Valid
31. Valid
32. Valid
33. Valid
34. Invalid — Lepton number
35. Invalid — Lepton number
36. Valid
37. Valid
38. Invalid — Charge
39. Invalid — Charge
40. Valid
41. Valid
42. Valid
43. Valid
44. Valid
45. Valid
46. Invalid — Lepton number
47. Valid
48. Valid
49. Valid
50. Valid